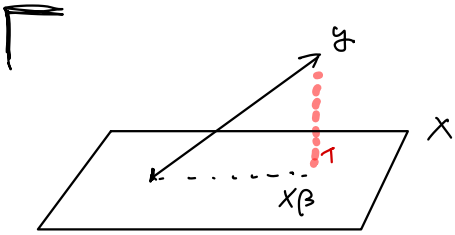


수선의 발.

$$X^T y = X^T X \beta$$

β를 어떻게 찾은가?

Least Square Estimate

$$(\text{loss}) = \|y - X\beta\|^2$$

$$= (y - X\beta)^T (y - X\beta)$$

$$= (y^T - \beta^T X^T)(y - X\beta)$$

$$= y^T y - y^T X\beta - \beta^T X^T y + \beta^T X^T X \beta$$

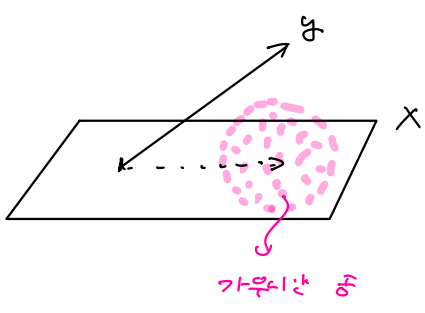
$$\nabla_\beta \beta^T A \beta = \nabla_\beta \sum_{i,j} \beta_i A_{ij} \beta_j$$

$$= (A + A^T)\beta$$

$$\nabla_\beta (\text{loss}) = -2X^T y + 2X^T X \beta = 0 \quad (\text{극값})$$

hence $X^T y = X^T X \beta$ 수선의 발

Maximum Likelyhood Estimate



let $Y = X\beta + \varepsilon \sim N(X\beta, \sigma^2 I)$

then $p(Y) = \dfrac{1}{(2\pi)^{d/2} \cdot \sigma^d} \cdot \exp\left(\dfrac{-1}{2\sigma^2} \|y - X\beta\|^2\right)$

likelyhood

squared L2 Norm

$p(\theta)$: prior

$p(\text{data}|\theta)$: likelyhood

$p(\text{data})$: evidence.

$p(\theta|\text{data})$: posterior $= \dfrac{p(\text{data}|\theta)\cdot p(\theta)}{\cancel{p(\text{data})}}$

Bernoulli R.V "동전 던지기"

$$p(x_i|\theta) = \begin{cases} \theta & x_i = 1 \\ 1-\theta & x_i = 0 \end{cases}$$

$$= \theta^{x_i}(1-\theta)^{1-x_i}$$

---

likelyhood 의 관점.

$$p(x|\theta) = \theta^{\sum x_i}(1-\theta)^{n-\sum x_i}$$

say $\nabla_\theta \log p(x|\theta) = 0$ (MLE)

$$\log p(x|\theta) = \sum x_i \log\theta + (n-\sum x_i)\log(1-\theta)$$

then .... $\theta = \dfrac{\sum x_i}{n}$

$$\nabla_\theta \log p(x|\theta) = \frac{\sum x_i}{\theta} - \frac{(n-\sum x_i)}{(1-\theta)}$$

$$\text{Beta}(\alpha,\beta) = x^{\alpha-1}\cdot(1-x)^{\beta-1}\cdot\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\cdot\Gamma(\beta)}$$

where $0 \le x \le 1$

Note.

$\Gamma(n) = (n-1)!$

---

posterior 의 관점

실험결과 H 7 T 3

$\nabla_\theta \log p(\theta|X) = 0$ (MAP)

Beta(1,1) Beta(8,4)

$$p(\theta|X) \propto \underbrace{1}_{p(\theta)}\cdot\underbrace{\theta^7(1-\theta)^3}_{p(X|\theta)} = \theta^7(1-\theta)^3$$

$\approx$ Uniform Distribution 이니까

$\theta_{MAP} = \theta_{MLE}$ ☆☆

$= 0.7$

Beta(2,2) Beta(9,5)

$$p(\theta|X) \propto \underbrace{\theta^1\cdot(1-\theta)^1}_{p(\theta)}\ \underbrace{\theta^7(1-\theta)^3}_{p(X|\theta)} = \theta^8(1-\theta)^4$$

$\approx \theta_{MAP} = 0.666\ldots$

엔트로피

- $h(x) = -\log p(x)$
- if x, y independent

$$h(x,y) = h(x) + h(y)$$

- $H(X) = \mathbb{E}[-\log p(X)] \geq 0$
- $H(X,Y) = H(X) + H(Y|X)$

(예)

$H \uparrow$ 주사위 vs 스페인

$H \Downarrow$ 주사위 vs 한국

KL Divergence

$$D_{KL}(p \| q)$$

$$= \sum_x p(x) \log \frac{p(x)}{q(x)}$$

$$= \int p(x) \log \frac{p(x)}{q(x)} dx \geq 0$$

(x) symmetric

(x) triangular inequality

(pf)

$$\mathbb{E}_{p(x)}\left[-\log \frac{q(x)}{p(x)}\right]$$

Const

$$\geq -\log \mathbb{E}_{p(x)}\left[\frac{q(x)}{p(x)}\right]$$

Const = 1

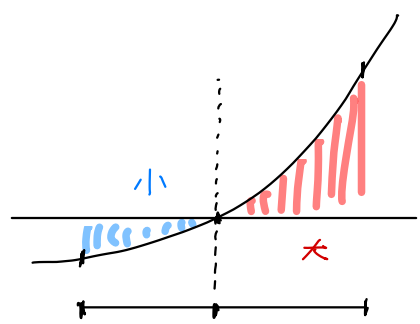
$$= -\log \sum_x \cancel{p(x)} \cdot \frac{q(x)}{\cancel{p(x)}} \geq 0$$

Jenson's Inequality

f convex

then $E[f(x)] \geq f(E(x))$

$f(x) = \text{Const}$



Mutual Information

$$I(X;Y) = D_{KL}(p(x,y) \| p(x) \cdot p(y))$$

$$= \sum p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$$

$$= H(X) + H(Y) - H(X,Y)$$

$$= H(X) - H(X|Y)$$

Cross Entropy

- $H_p(q) = \mathbb{E}_p[-\log q(x)]$

$$= -\sum_x p(x) \log q(x)$$

- $D_{KL}(p \| q) = \mathbb{E}_p\left[\log \frac{p(x)}{q(x)}\right]$

$$= H_p(q) - \cancel{H(p)} \geq 0$$

fixed

overfitting

$$y = \underbrace{\begin{bmatrix} 1 & x^1 & \cdots & x^m \end{bmatrix}}_{\text{단일 변수에 의존}} \underbrace{\begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{bmatrix}}_{\text{불필요하게 큰 차원}} \Rightarrow \text{overfitting}$$

Regularization

$$(\text{loss}) = \|Y - X\beta\|^2 + \underbrace{\lambda\|\beta\|^2}_{\text{Regularization}}$$

$$\nabla_\beta(\text{loss}) = \nabla_\beta\left(y^Ty - y^TX\beta - \beta^TX^Ty + \beta^TX^TX\beta\right) + \nabla_\beta \lambda\beta^T\beta$$

$$= 2\left[\underbrace{-X^T(Y - X\beta)}_{\text{레이블 예측}} + \underbrace{\lambda\beta}_{\text{단순한 모델}}\right] = 0$$

Kernel

$$K(\underbrace{x_1, x_2}_{\text{low dimension}}) = \phi(x_1)^T \underbrace{\phi(x_2)}_{\text{high dimension}}$$

$$(\text{loss}) = \|Y - \phi(X)^T\beta\|^2 + \lambda\|\beta\|^2$$

$$= y^Ty - y^T\phi(X)^T\beta - \beta^T\phi(X)Y + \beta^T\phi(X)\phi(X)^T\beta + \lambda\beta^T\beta$$

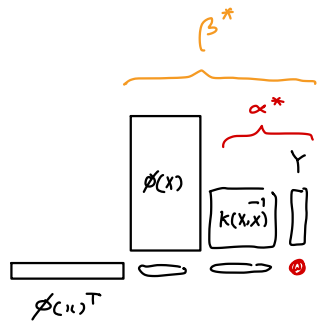
$$\nabla_\beta(\text{loss}) = 2\left[-\phi(X)(Y - \phi(X)^T\beta) + \lambda\beta\right] = 0$$

$$\beta^* = (\phi(X)\phi(X)^T + \lambda I)^{-1}\phi(X)Y$$

$$= \phi(X)\underbrace{(\phi(X)^T\phi(X) + \lambda I)^{-1}Y}_{=\alpha^*}$$

$$f^*(x) = \phi(x)^T\beta^*$$

$$= K(x, X)\alpha^*$$

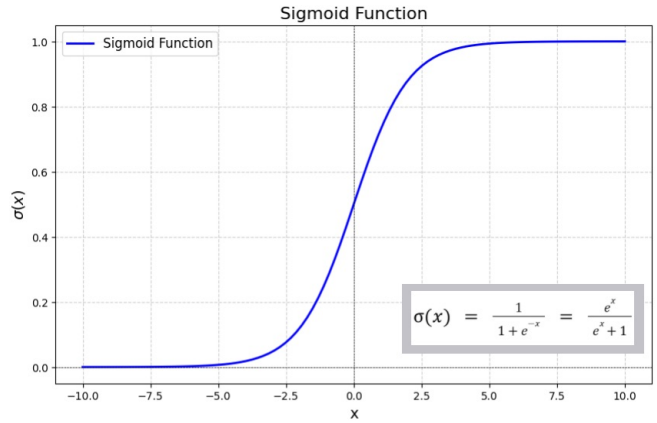


sigmoid 임의의 값을 확률 형태로 변환해줌.

if probability $= \dfrac{e^z}{1+e^z}$ $[0, 1]$

then odds $= e^z$ $[0, \infty)$

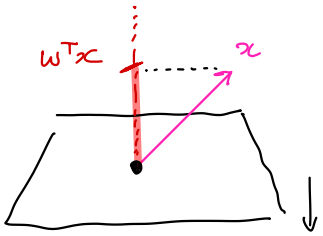
$\log(\text{odds}) = z$ $(-\infty, \infty)$



Note that $\sigma' = \sigma(1-\sigma)$

Logistic Regression (MLE)

$$\sigma' = \sigma(1-\sigma)$$



아래로 당기면
신뢰도 상승

$$\nabla_w \sigma_i = \sigma_i(1-\sigma_i)\,\nabla_w(w^T x_i)$$
$$= \sigma_i(1-\sigma_i)\,x_i$$
$$\propto x$$

$$p(y_i \mid x_i, w) = \left(\sigma(w^T x_i)\right)^{y_i}\left(1-\sigma(w^T x_i)\right)^{1-y_i}$$

$$\mathcal{L} = \prod_{i=1}^{N} p_i$$

data

$$\log p_i = \left(y_i \log \sigma_i\right) + \left((1-y_i)\log(1-\sigma_i)\right)$$

$$\nabla_w(-\log p_i) = \left(y_i(\sigma_i - 1)\,x_i\right) + \left((1-y_i)\,\sigma_i\,x_i\right)$$

$$= \left(y_i\sigma_i x_i - y_i x_i\right) + \left(\sigma_i x_i - y_i \sigma_i x_i\right)$$

$$= (\underbrace{\sigma_i}_{\text{예측}} - \underbrace{y_i}_{\text{실제}})\,x_i$$

예측 실제 피처벡터 위치

hence $\nabla_w(-\log \mathcal{L}) = \sum_i \left(\overbrace{\sigma(w^T x_i)}^{\text{예측}} - \overbrace{y_i}^{\text{실제}}\right) \overbrace{x_i}^{\text{피처벡터 위치}}$

$$\nabla_w \nabla_w(-\log p_i) = \left[\nabla_w(\sigma_i - y_i)\right] x_i^T$$
$$= \sigma_i' x_i x_i^T$$

for any v

$$v^T(X^T S X)v$$
$$= (Xv)^T S (Xv)$$
$$= \sum_i \underbrace{(\sigma_i')}_{\geq 0}\underbrace{\left((Xv)_i\right)^2}_{\geq 0} \geq 0$$

$$\nabla_w \nabla_w(-\log \mathcal{L}) = \underbrace{\begin{bmatrix} | & \\ x_1 & \cdots \\ | & \end{bmatrix}}_{X^T}\underbrace{\begin{bmatrix} \sigma_1' & & \\ & \ddots & \\ & & \sigma_n' \end{bmatrix}}_{S}\underbrace{\begin{bmatrix} - & x_1 & - \\ & \ddots & \end{bmatrix}}_{X}$$

$\sigma_1' \geq 0$, $\sigma_n' \geq 0$

because $\sigma' = \sigma(1-\sigma) \geq 0$

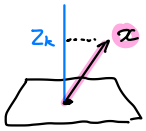
positive semi-definite $\longleftrightarrow$ $-\log \mathcal{L}$ convex

$\langle v, Hv \rangle \geq 0$

有 global min

Multiclass classification

let $z_k = w_k^T x$



$$\nabla_{w_m}(w_k^T x) = \begin{cases} x & \text{if } m=k \\ 0 & \text{if } m \neq k. \end{cases}$$

$$= \delta_{mk} \cdot x$$

then $p_k = \dfrac{e^{z_k}}{\sum_j e^{z_j}}$

$$\nabla_{w_m}(p_k) = \frac{\sum_j e^{z_j}\left(\overset{\delta_{mk}\cdot x}{e^{z_k}\cdot(\nabla z_k)}\right) - e^{z_k}\left(\overset{e^{z_m}\cdot x}{\sum_j e^{z_j}\nabla z_j}\right)}{\left(\sum_j e^{z_j}\right)^2}$$

$$= p_k(\delta_{mk} - p_m)\, x$$

$$\nabla_{w_m}\left(-\log \begin{matrix}\text{likelyhood}\\ \text{of } x\end{matrix}\right) = \sum_{k=1}^{c} y_k \nabla(-\log p_k)$$

$$= \sum_{k=1}^{c} y_k \cdot \frac{1}{\cancel{p_k}} \cdot \cancel{p_k}(p_m - \delta_{mk})\, x$$

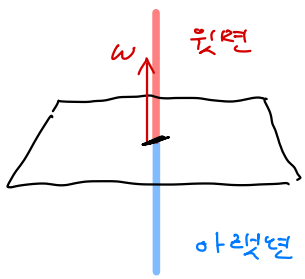
$$= (p_m - y_m)\, x$$

$$\nabla_{w_m}(-\log \mathcal{L}) = \sum_{i=1}^{N} (\underset{\text{cal}(\hat{y}_i)}{p_{im}} - \underset{\text{실제}}{y_{im}})\, x_i$$

SVM 초평면 $w^T x + b = 0$

조건 ①

$$\forall i \quad y_i (w^T x_i + b) \geq 0$$



$y = 1$

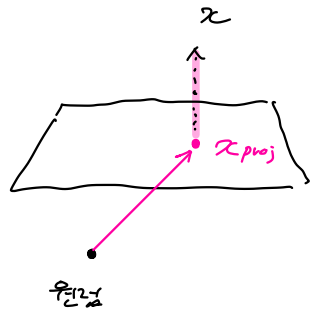
$w^T x + b \geq 0$

$w^T x + b \leq 0$

$y = -1$

조건 ②

$$\min_x |w^T x + b| = 1$$



$$x = x_{proj} + d \frac{w}{\|w\|}$$

$$d = \frac{|w^T x + b|}{\|w\|}$$

$$= \frac{1}{\|w\|}$$ (가장 가까운 점)

$\frac{1}{2}\|w\|^2$ $\qquad$ $y_i (w^T x_i + b) \geq 1$

$$\text{minimize} \quad \frac{1}{2}\|w\|^2 + \max_{\alpha_i \geq 0} \alpha_i \left(1 - y_i (w^T x_i + b)\right) = \begin{cases} 0 & y_i (w^T x_i + b) > 1 \\ \infty & \text{else} \end{cases}$$

$$\min_{w,b} \max_{\alpha_i \geq 0} \left[\frac{1}{2}\|w\|^2 + \sum_i \alpha_i (1 - y_i (w^T x_i + b)) \right] \geq \max_{\alpha_i \geq 0} \min_{w,b} \left[\quad '' \quad \right]$$

$$= \frac{1}{2} w^T w + \sum_i \alpha_i - w^T \underbrace{\left(\sum_i \alpha_i y_i x_i\right)}_{w} - b \underbrace{\sum_i \alpha_i y_i}_{0}$$

maximize

$$\sum_i \alpha_i - \frac{1}{2} w^T w$$

with

$$\begin{cases} \nabla_w \textcircled{1} = w - \sum_i \alpha_i y_i x_i = 0 \\ \frac{\partial}{\partial b} \textcircled{2} = \sum_i \alpha_i y_i = 0 \end{cases}$$

say $i \in I$ whenever $\alpha_i > 0$

i.e $y_i(w^T x_i + b) = 1$

then $w = \sum_i \alpha_i y_i x_i \quad (i \in I)$

$b = y_i - w^T x_i \quad (i \in I)$

$$w^T w = \sum_i \alpha_i y_i (w^T x_i) \quad (i \in I)$$

$$= \sum_i \alpha_i (1 - y_i b) \quad \text{because} \quad y_i(w^T x_i + b) = 1$$

$$= \sum_i \alpha_i - b \cancel{\sum_i \alpha_i y_i}$$

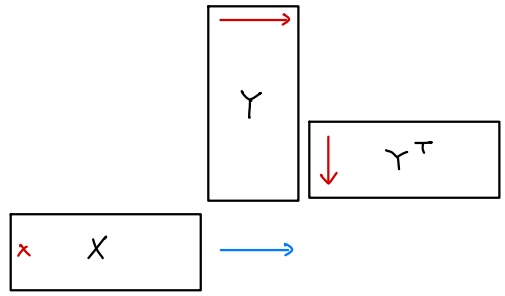
$$\frac{|x^T w + b|}{\|w\|} = \frac{1}{\sqrt{\sum \alpha_i}}$$

---

say $W = XY$

then $\frac{\partial \mathcal{L}}{\partial X} = \frac{\partial \mathcal{L}}{\partial W} \cdot Y^T$

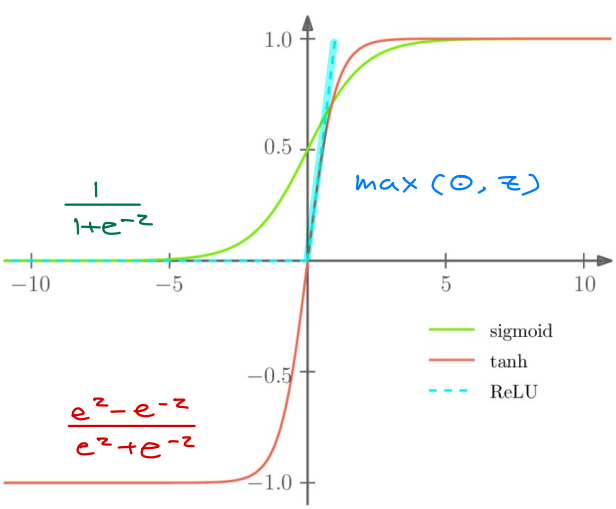
$\frac{\partial \mathcal{L}}{\partial Y} = X^T \frac{\partial \mathcal{L}}{\partial W}$



Gradient Descent $\quad \theta_{k+1} = \theta_k - \alpha \nabla \mathcal{L}(\theta_k)$

minimize

- Full Batch GD $\quad \tilde{\nabla} f(x) = \frac{1}{N} \sum_{i=1}^{N} \nabla f_i(x)$
- Stocastic GD $\quad \tilde{\nabla} f(x) = \nabla f_i(x)$
- Mini Batch GD



$\sigma' = \sigma(1-\sigma)$

$(\tanh)' = 1 - (\tanh)^2$

$(\mathrm{ReLU})' = \begin{cases} 1 & z > 0 \\ 0 & \end{cases}$

Note. σ나 tanh는 (대부분) linear

let $z = \sum_{i=1}^{D_{in}} w_i x_i$ s.t $\begin{cases} w : \text{i.i.d}, \ E[w_i] = 0 \quad Var[w_i] = \sigma^2 \\ x : \text{i.i.d independent of } w \\ \quad E[x_i] \neq 0 \ (\text{in general}) \end{cases}$

let $S = E[x^2]$ denote the 2nd moment of the input.

---

we know that $E[z] = 0$

Note. independent

$$\iint xy \, p(x,y) \, dx \, dy$$

$$= \iint xy \, p(x) p(y) \, dx \, dy$$

then $Var(z) = \sum_{i=1}^{D_{in}} Var(w_i x_i)$

$$= \sum_{i=1}^{D_{in}} E[w_i^2 x_i^2] - E[w_i x_i]^2 = D_{in} \sigma^2 S$$

$$E[\cancel{w_i^2}] E[\cancel{x_i^2}] \qquad E[\cancel{w_i}]^2 E[x_i]^2$$

$$\sigma^2 \quad S \qquad 0$$

in Xavier Init we want $E[z^2] = E[x^2]$

$$\cancel{D_{in} \sigma^2 S} = \cancel{S} \qquad \sigma = \frac{1}{\sqrt{D_{in}}}$$

---

now, let $h = ReLU(z)$

then $E[h^2] = \int_0^\infty z^2 p(z) \, dz$

$$= \frac{1}{2} \int_{-\infty}^{\infty} z^2 p(z) \, dz = \frac{1}{2} E[z^2]$$

$$= \frac{1}{2} D_{in} \sigma^2 S$$

in He Init we want $E[h^2] = E[x^2]$

$$\cancel{\frac{1}{2} D_{in} \sigma^2 S} = \cancel{S} \qquad \sigma = \sqrt{\frac{2}{D_{in}}}$$

$$\begin{cases} \text{Xavier} \quad \sigma^2_{reg} = \frac{1}{D_{in}} \\ \text{He} \quad \sigma^2_{reg} = \frac{2}{D_{in}} \end{cases}$$

then $w \sim N(0, \sigma^2_{reg})$

or $w \sim Uni(-\sqrt{3}\sigma_{reg}, \sqrt{3}\sigma_{reg})$

as variance of $Uni(-b, b)$ becomes

$$\int_{-b}^{b} x^2 \cdot \frac{1}{2b} \, dx$$

$$= \left[\frac{x^3}{6b} \right]_{-b}^{b} = \frac{b^3}{3}$$

learning rate decay

① at a few fixed points ?

② cosine : $\frac{1}{2}\alpha_0\left(1+\cos\left(\frac{t\pi}{T}\right)\right)$

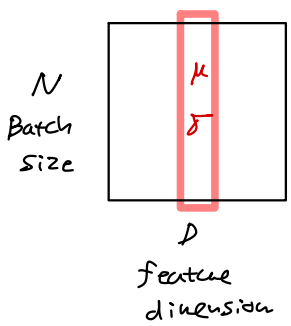
③ linear : $\alpha_0\left(1-\frac{t}{T}\right)$

④ linear sqrt : $\alpha_0/\sqrt{t}$

linear warmup

: linearly increase learning steps at $t\pi$ ☺

Batch Normalization



$$\tilde{x}_{ij} = \frac{x_{ij} - \mu_j}{\sqrt{\sigma_j^2 + \varepsilon}}$$

$$\tilde{y}_{ij} = \gamma_j \tilde{x}_{ij} + \beta_j$$

(training)

$$\mu^{run} = m \cdot \mu^{run} + (1-m)\,\mu^{batch}$$

$$(\sigma^2)^{run} = m(\sigma^2)^{run} + (1-m)(\sigma^2)^{batch}$$

BN $\not\Rightarrow$ Internal Convergence Shift : shift in the mean/var of hidden activation

$\Rightarrow$ Smooth loss landscape : stable gradient.

GD minimizes $f(x)$

$$x_{t+1} = x_t + \eta u$$

to minimize $f(x_{t+1}) \approx f(x_t) + \langle \nabla f(x_t), \eta u \rangle$

set $u = -\dfrac{\nabla f(x_t)}{\| f(x_t) \|}$ GD idea

Lemma 3.1

- $f: \mathbb{R}^d \to \mathbb{R}$ continuously diffable function.

- let f β-smooth : $|\nabla f(x) - \nabla f(y)| \le \beta \|x-y\|$ $(\forall x, y)$

then $f(y) \le f(x) + \underbrace{\langle f(x), y-x \rangle}_{\text{한단}} + \underbrace{\frac{\beta}{2} \|y-x\|^2}_{\text{도함수}}$

GD : $x_{t+1} = x_t - \eta \nabla f(x_t)$

$$f(x_{t+1}) \le f(x_t) + \langle \nabla f(x_t), -\eta \nabla f(x_t) \rangle + \frac{\beta}{2} \|\eta \nabla f(x_t)\|^2$$

$$= (\cdots) = f(x_t) - \underbrace{\left(\eta - \frac{\beta}{2}\eta^2\right)}_{\text{small step}} \|\nabla f(x_t)\|^2$$

SGD : $x_{t+1} = x_t - \eta \tilde{\nabla} f(x_t)$

then $f(x_{t+1}) \le f(x_t) - \eta \langle \nabla f(x_t), \tilde{\nabla} f(x_t) \rangle + \frac{\beta}{2} \cdot \eta^2 \|\tilde{\nabla} f(x_t)\|^2$

$$E_t[f(x_{t+1})] \le f(x_t) - \eta \|\nabla f(x_t)\|^2 + \frac{\beta}{2} \eta^2 \underbrace{E_t[\|\tilde{\nabla} f(x_t)\|^2]}_{=:G}$$

$$E[\|\nabla f(x_t)\|^2] \le \frac{1}{\eta}\left(E[f(x_t)] - E[f(x_{t+1})]\right) + \frac{\beta}{2} \eta^2 G$$

$$\sum_{t=0}^{T-1} E[\|\nabla f(x_t)\|^2] \le \frac{1}{\eta}(f_0 - \overset{\text{lower bound}}{f^*}) + \frac{\beta}{2} \eta^2 G T$$

$$\min_t \left(E\|\nabla f(x_t)\|^2\right) \le \frac{1}{\eta T}(f_0 - f^*) + \frac{\beta}{2} \eta^2 G$$

let $\eta = \frac{1}{\sqrt{T}}$

then $\min(E[\|\nabla f(x_t)\|^2]) = O(\frac{1}{\sqrt{T}})$

$$g(t) = f\left(x + t(y-x)\right)$$

$$g'(t) = (y-x)^T \nabla f(x + t(y-x))$$

$$g''(t) = (y-x)^T \nabla^2 f(x+t(y-x))\,(y-x) \quad \leq \beta \|y-x\|^2$$

$$\int_0^1 (1-s)\, g''(s)\, ds = \Big[(1-s)\, g'(s)\Big]_0^1 + \int_0^1 g'(s)\, ds$$

$$= g(1) - g(0) - g'(0)$$

Hence $f(y) = f(x) + \langle \nabla f(x), y-x \rangle + \underline{\int_0^1 (1-s)(y-x)^T \nabla^2 f(x+s(y-x))\,(y-x)\, ds}$

$$\leq \int_0^1 (1-s)\, \beta \|y-x\|^2\, ds$$

$$= \frac{\beta}{2} \|y-x\|^2$$

---

let $\phi(t) = \nabla f(x + t(y-x))$

$$\phi'(t) = \nabla^2 f(x+t(y-x))\,(y-x)$$

then $\nabla f(y) - \nabla f(x) = \int_0^1 \phi'(t)\, dt$

$$= \int_0^1 \nabla^2 f(x+t(y-x))\, dt \cdot (y-x)$$

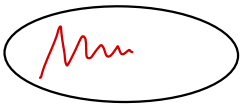
$$(y-x)^T \int_0^1 \nabla^2 f(x+t(\cancel{y}-x))\, dt\, (y-x)$$

$$= \langle \nabla f(y) - \nabla f(x), y-x \rangle \leq \|\nabla f(y) - \nabla f(x)\|\, \|y-x\|$$

$$\leq \beta \|y-x\|^2$$

GD

$$x_{t+1} = x_t - \alpha \nabla f(x_t)$$



항복운동

+Momentum

$$x_{t+1} = x_t + v_{t+1}$$

$$v_{t+1} = \rho v_t - \alpha \nabla f(x_t)$$

Nesterov Momentum

$$x_{t+1} = x_t + v_{t+1}$$

$$v_{t+1} = \rho v_t - \alpha \nabla f(x_t + \rho v_t)$$

$\| \nabla f(x) \|$. 가파른 정도 .

" 가파르면 느리게

완만하면 빠르게 "

AdaGrad

$$x = x - \alpha \frac{\nabla f(x_t)}{\sqrt{\sum (\nabla f(x_t))^2}}$$

$\rightarrow \infty$

RMSProp

$$x = x - \frac{\eta g_t}{\sqrt{E[g^2]_t}}$$

$$E[g^2]_t = \beta E[g^2]_{t-1} + (1-\beta) g_t^2$$

Adam

$$x_{t+1} = x - \alpha \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t}}$$

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t \qquad m_0 = 0$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t \qquad v_0 = 0$$

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t} \qquad \hat{v}_t = \frac{v_t}{1-\beta_2^t}$$

---

population risk $\quad R(f) = \mathop{\mathbb{E}}_{xy}\left[\ell(f(x), y)\right]$

Note.

$$R(f) = \mathbb{E}[\hat{R}(f)]$$

empirical risk $\quad \hat{R}(f) = \frac{1}{N}\sum_{i=1}^{N} \ell(f(x_i), y_i)$

Approximation

Hypothesis Space $\quad$ complexity measure

let $\quad F_\delta := \{ f \in F : r(f) \le \delta \}$

suppose

F normed space.

Risk Decomposition

generalization

$$\underbrace{R(\hat{f}) - \inf_{f \in F} R(f)}_{\text{total risk}} = R(\hat{f}) - \hat{R}(\hat{f})$$

optimization

$$+ \hat{R}(\hat{f}) - \inf_{F_\delta} \hat{R}(f)$$

$$+ \inf_{F_\delta} \hat{R}(f) - \inf_{F_\delta} R(f)$$

approximation

$$+ \inf_{F_\delta} R(f) - \inf_{F} R(f)$$

$$\le 2 \cdot \sup_{f \in F_\delta} |R(f) - \hat{R}(f)| + (\text{opt}) + (\text{apx})$$

$\because$ $\quad \inf_{F_\delta} \hat{R}(f) - \inf_{F_\delta} R(f^*)$

$$\le \hat{R}(f^*) - R(f^*)$$

$$\le \sup_{F_\delta} |R(f) - \hat{R}(f)|$$

$\quad \inf_{F_\delta} R(f) - \inf_{F_\delta} \hat{R}(f^*)$

$$\le R(f^*) - \hat{R}(f^*)$$

$$\le \sup_{F_\delta} |R(f) - \hat{R}(f)|$$

Generalization Error

and the curse of dimensionality

- $\mathbb{E}[\hat{R}(f)] = R(f)$

- $\mathrm{Var}[\hat{R}(f) - \underbrace{\cancel{R(f)}}_{\text{const}}] = \frac{1}{n^2} \cdot n \, \mathrm{Var}[\ell(f(x), y)]$

$\propto \frac{1}{n}$

---

set $f^* = \langle x, \theta^* \rangle$

$\hat{f} = \langle x, \hat{\theta} \rangle$ (train)

$\theta^*, \hat{\theta} \in \mathbb{R}^d$

hence $n \geq d$

minimize $\mathbb{E}\left[|f^*(x) - \overset{\text{train}}{\hat{f}(x)}|\right] \leq \varepsilon$

---

f^* is $\begin{cases} \text{locally linear} \\ |f^*(x_1) - f^*(x_2)| \leq \beta |x_1 - x_2| \quad \text{(Lipschitz)} \end{cases}$

$\hat{f} = \underset{f}{\operatorname{argmin}} \{ \mathrm{Lip}(f) : f^*(x_i) = f(x_i) \}$ 립시츠 상수가 가장 낮은 녀석

then $|f^*(x) - \hat{f}(x)| \leq |f^*(x) - f^*(x_i)| \leq \mathrm{Lip}(\hat{f}) \overbrace{\|x - x_i\|}^{\text{가장 가까운 샘플}} + \beta \|x - x_i\|$

$+ \cancel{|f^*(x_i) - \hat{f}(x_i)|}$

$+ |\hat{f}(x_i) - \hat{f}(x_i)|$

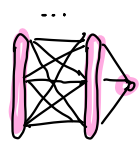
---

unfortunately $\|x - x_i\| \approx \frac{1}{n^{1/d}} \leq \varepsilon$ requires $n \approx \left(\frac{1}{\varepsilon}\right)^d$

$\underbrace{n}_{\text{개수}} \cdot \underbrace{\ell^d}_{\text{부피}} = 1$

"데이터가 고차원 속이 low-dimensional Manifold에 갇혀 있는 이유"

Universal Approximation Thm.



$$G(x) = \sum_{j=1}^{N} \alpha_j \, \sigma(w_j^T x + \theta_j)$$

are dense in $C(I_n)$

$\sigma : \mathbb{R} \to [0,1]$

is called sigmoid

if ① $\sigma \in C^1(\mathbb{R})$

② $\sigma(+\infty) = 1$

③ $\sigma(-\infty) = 0$

Lemma 1.

any conti sigmoid func is discriminatory for $\mu \in \mathcal{M}(I_n)$

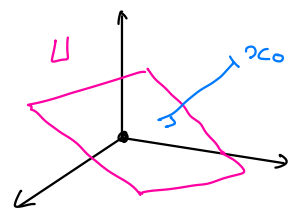
i.e if $\int_{I_n} \sigma(w^T x + \theta) \, d\mu(x) = 0$

for $\forall w \in \mathbb{R}^n$, $\forall \theta \in \mathbb{R}$ then $\mu = 0$

Lemma 2.

① U : linear subspace of normed linear space X

② U is not dense $\left(\begin{array}{l} \exists x_0 \in X \\ \text{s.t } d(x_0, U) \geq \delta > 0 \end{array} \right)$



then 有 bounded linear func L on X

s.t (i) $L|_U = 0$ i.e $L(u) = 0 \quad \forall u \in U$

(ii) $L \neq 0$ $\begin{cases} L(x_0) = \delta \\ \|L\| < 1 \end{cases}$

Note. bounded linear func.

$$\|L\| = \sup_{x \neq 0} \frac{|L(x)|}{\|x\|}$$

Thm. RRT

① compact $K \subseteq \mathbb{R}^n$

② $C(K)$: set of real valued conti func on K

③ F : bounded linear func on $C(K)$

then 有 unique finite signed measure μ on K

$$F(f) = \int_K f(x) \, d\mu(x) \qquad \forall f \in C(K)$$

Lemma 3.

$\mathcal{U}$: linear, non-dense subspace of $C(I_n)$

then $\exists \mu \in \mathcal{M}(I_n)$

s.t $\int_{I_n} h \, d\mu = 0 \quad \forall h \in \mathcal{U}$

$\because$ $\exists$ bounded linear func L on $C(I_n)$

$\exists \mu \in \mathcal{M}(I_n)$

s.t $L(f) = \int_{I_n} f \, d\mu(x) \quad \forall f \in C(I_n)$

$$\begin{cases} L|_{\mathcal{U}} = 0 & : L(h) = 0 \quad \forall h \in \mathcal{U} \\ L \neq 0 & : \mu \neq 0 \end{cases}$$

Lemma 4.

σ be any conti discriminatory func.

then $\mathcal{U} = \{ G : G(x) = \sum_{j=1}^{N} \alpha_j \, \sigma(w_j^T x + \theta_j) \}$ linear subspace of $C(I_n)$

are dense in $C(I_n)$

$\because$ suppose $\mathcal{U}$ not dense

then $\exists \mu \neq 0$ s.t $\int_{I_n} h \, d\mu = 0 \quad (\forall h \in \mathcal{U})$

$$\sum_j \alpha_j \int_{I_n} \sigma(w_j^T x + \theta_j) \, d\mu = 0 \qquad \forall w_j \in \mathbb{R}^n, \ \forall \alpha_j, \theta_j \in \mathbb{R}$$

$$\int_{I_n} \sigma(w_j^T x + \theta_j) \, d\mu = 0$$

since σ discriminatory $\mu = 0$

Rademacher Complexity

- $P(\sigma_i = 1) = P(\sigma_i = -1) = \frac{1}{2}$

- $\mathrm{Rad}_n(\mathcal{F}) = E\left[\sup_{f \in \mathcal{F}} \left(\frac{1}{n} \sum_{i=1}^{n} \underbrace{\sigma_i f(z_i)}_{\text{노이즈를 따라가는 정도.}} \right)\right] \propto$ 과적합 위험

---

let $\mathcal{F}_{m,\sigma} = \left\{ f_\theta(x) = \sum_{j=1}^{m} \beta_j \, \sigma(w_j^T x) \right\}$

then $\mathrm{Rad}(\mathcal{F}_{m,\sigma}, a) \lesssim \frac{a}{\sqrt{n}}$

$\leftarrow$ 리라티리 크기

$\leftarrow$ 레이러 샘플 갯수

정리 sigmoid σ

$f \in P_{C_1}$, $c > 4C_1^2$ $N \in \mathbb{N}$

then $\exists \Phi_N$ s.t $\frac{1}{|B_1^d|} \int_{B_1^d} |f(x) - \Phi(x)|^2 dx \leq \frac{c}{N}$